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With support from AWS, Bravante RRC uses generative AI to find deactivated oil pipelines

[Bravante RRC](#) relies on support from AWS and Flexa Cloud in its biggest challenges, such as identifying oil equipment that must be removed from the bottom of the sea.

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Create

a single source of data on the stretches

Reduction

of planning errors

Man hours

savings, reducing the burden on survey

Reduction

of decommissioning errors, at a cost of US\$ 250 thousand each



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Overview

[RRC Tecnologia](#) is the robotics and technology arm of [Grupo Bravante](#). Created in 2012, it is also known as Bravante RRC. Its teams faced one of the greatest challenges in marine engineering, the so-called decommissioning of oil equipment and the removal of lines and pipelines (stretches) from the bottom of the sea that are no longer being used for petroleum exploration.



Opportunity | Preserving the Seabed

The process is regulated by the National Petroleum Agency, in resolution 817/2020, which states that the seabed must be left as it was found after exploration. Felipe Santello, product manager at RRC, explains that since there are pipelines that have been there for over 30 years, finding them depends on analyzing documents that are often scattered and in different formats. It is a task carried out by naval specialists who analyze them and determine where this equipment should be found. The issue is that the locations are not always accurate and the daily rate of the ship carrying out the certification is US\$ 250k, which implies a loss if the equipment is not actually found and removed.

"The purpose of the vessel is to verify that the pipeline is in its documented state, verifying the weight, condition of the stretch and its existing marine life, which also needs to be preserved," he explains. "To give you an idea, each stretch has six or seven reports - 15 to 40 pages each - in addition to 2D and 3D images that need to be read and analyzed. Considering the more than 800 km of pipelines to be decommissioned, the data repository has over 10 thousand documents."

To reduce the manual work involved in the analysis of such a complex repository and increase the accuracy of the locations, the company decided to develop a solution based on Generative Artificial Intelligence that reads documents automatically and determine the exact location of the parts that must be removed.



has been increasingly required by the ANP and there is a need to gather all information in a single trustworthy source. We saw the existing potential.”

Why AWS | Analyzing Data in the Cloud

Amazon Web Services, which had been investing in the relationship with Grupo Bravante for months, was interested in the challenge and offered the company its support to develop and test an artificial intelligence (AI) platform. [Flexa Cloud](#), an official AWS partner with experience in corporate generative AI projects, was invited to build this solution with Bravante RRC scientists.

The architecture built starts with the creation of the document repository in [Amazon S3](#), allowing fast, secure, and low-cost access to thousands of documents. With native S3 object integration, [Amazon Textract](#) was used for optical character reading (OCR). The versatility of the service was crucial considering that the pipeline document repository includes images, spreadsheets, pdfs, and other formats.

“After extracting this data, we used the semantic capacity of a foundational model to interpret and classify the data at scale,” explains Flexa Cloud CEO, Deivid Bitti. “We started off with models that were customized using [Amazon SageMaker](#), which brought us a lot of computing power. Once we had access, we switched over to [Amazon Bedrock](#) architecture.”

In the view of Deivid Bitti, “Bedrock brings great versatility in the choice of foundational models. It facilitated the development and experimentation process by being quickly available via APIs.” Felipe Santello adds that “by using advanced interaction techniques with the foundational model, we achieve high precision without the need for prior training”.

The foundational model chosen was Anthropic's Claude-2. “The context window was a very important differential. We need to keep the model's attention for each stretch, which involves up to 7 reports with up to 40 pages each. The 100k token window was critical to our objective,” says Felipe.

Data extraction and processing via Amazon Bedrock allowed for the creation of a complete dossier on each of the stretches in an [Amazon Aurora](#)-structured database (DB). Bitti reveals that, with the data provided, they developed an preliminary platform from which Bravante could use and validate the solution. “We are testing a technology that can describe images in documents, in a scalable way, and enrich the Bravante's naval engineers,” concludes Deivid Bitti.

Results | Reducing Errors and Centralizing Data

According to Santello, the main expectation with the solution's development is that his team will have a single data source on the stretches. "This source will provide the chronological order and the current state of these stretches, defining which vessel and procedures to use in their removal, in addition to identifying the state of the surrounding marine life. These are the main questions we will ask this database (DB)," he explains.

The executive recalls that, in three months of the survey, about 100 stretches were identified, representing only a few of the thousands that must be identified and removed from the Brazilian coast. "It is important to highlight the size of the fields and the extent, in miles, of the lines that must be removed, so the most important part of the project is the knowledge of the people involved," he says.

In practice, the platform co-developed with Flexa and AWS brings gains such as reducing the operational load on surveying engineers, man hours savings, and avoiding errors. The increase in mapping accuracy will reduce the number of daily vessels required for field certification, which cost \$250,000 each. A reduction of up to 40% of these daily rates is expected over the course of the project, saving millions of dollars with the AWS solution.

About RRC Tecnologia and Grupo Bravante

Founded in 2012, RRC Tecnologia e Inovação is the robotics and technology arm of Grupo Bravante, producing innovative digital solutions for the oil & gas market and committed to market development needs.

AWS Services

Amazon Bedrock

Amazon Bedrock is a fully managed service that offers several high performance foundation model (FMs) options from the leading AI companies, such as AI21 Labs, Anthropic, Cohere, Meta, Stability AI and Amazon, through a single API, as well as a wide range of resources needed to build generative AI applications, simplifying development and maintaining privacy and security.

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Amazon Aurora

Amazon Aurora offers integrated security, continuous backups, serverless computing, up to 15 read replicas, multi-regional automated replication and integrations with other AWS products.

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Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service that offers industry-leading scalability, data availability, security, and performance.

[Learn more »](#)

Amazon Textract

Amazon Textract is a machine learning (ML) service that automatically extracts printed or handwritten text, layout elements, and data from scanned documents.

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